

Epsilon

Entity Relationship Diagram

Second-Hand Book / Gadget Exchange for Students

Database: epsilon_db 14 entities Normalised to Third Normal Form (3NF)

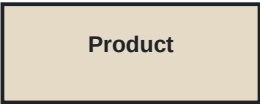
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3. Overview - all entities and how they relate
4. Accounts, categories and listings
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Generated 13 August 2026

Notation

The diagram uses Chen notation. Every shape below means the same thing on every page.



Entity

A table in the database.



Attribute

A column of that table.



Primary key

SOLID underline, bold black. Identifies one row uniquely.



Foreign key

DASHED underline, italic grey. Points to another table.



Unique key

Double outline. No two rows may repeat this value.



Relationship

How two entities are linked.



Cardinality

1 means one row, N means many rows.

Telling a key apart at a glance



solid line = primary key



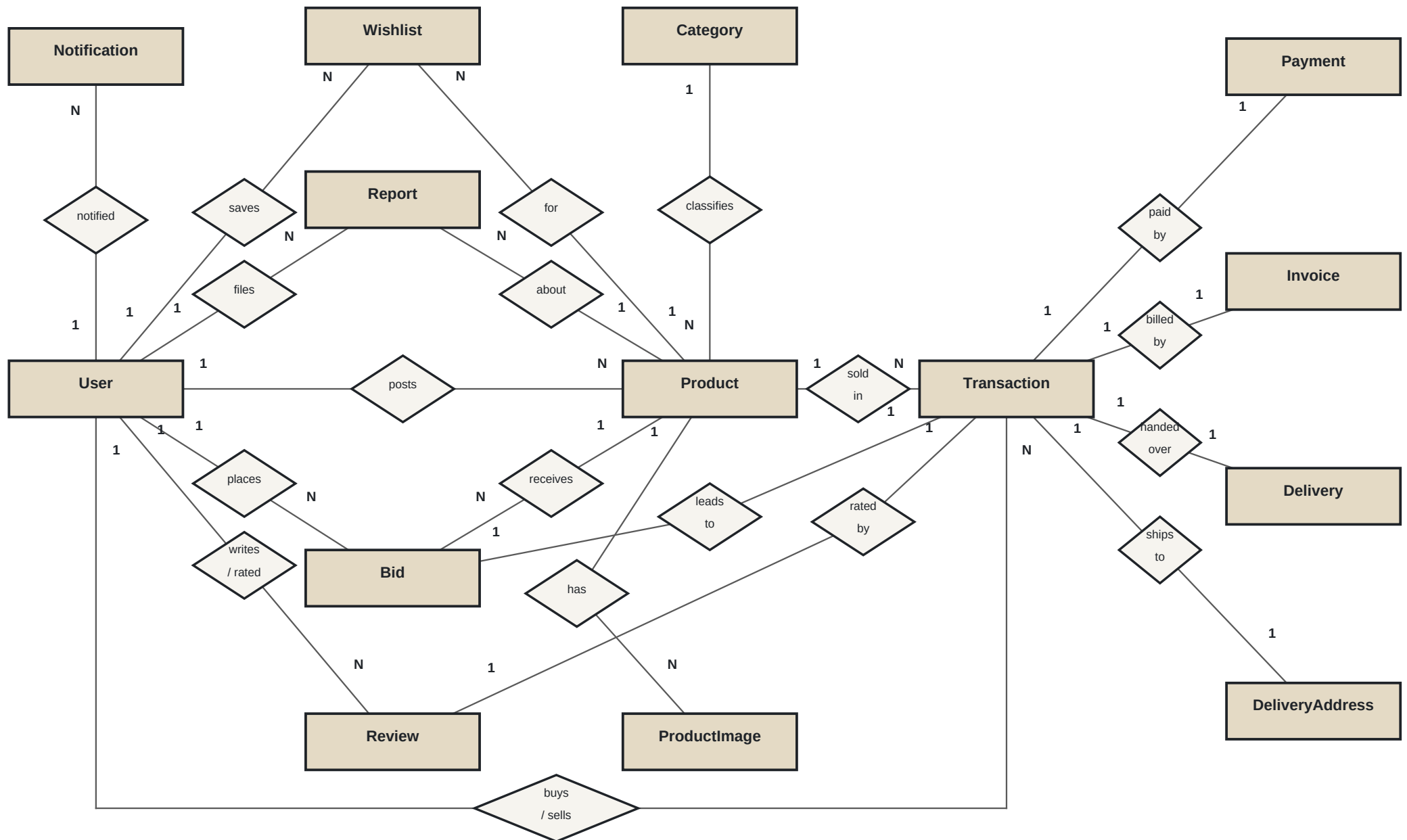
dashed line = foreign key

Both examples come from the Notification table. It has exactly one primary key, notification_id. product_id and transaction_id are foreign keys: they may be empty, and they record which item the notification is about so that clicking it opens the right page.

Strict Chen notation leaves foreign keys out, because the relationship diamond already carries that meaning. They are drawn here anyway - in italic grey with a dashed underline - so that every column in the database appears somewhere in this document. The last page lists all 23 of them in one table.

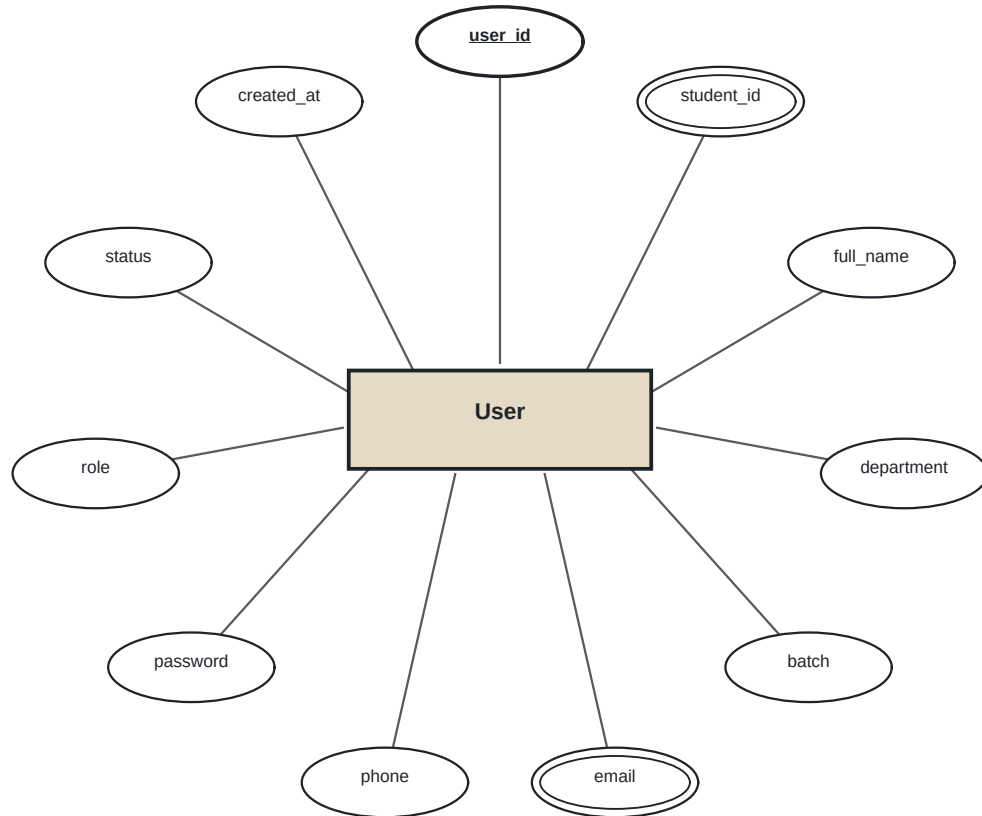
Overview - all 14 entities

Attributes are left out here so the relationships stay readable. Every entity is drawn in full on the pages that follow.



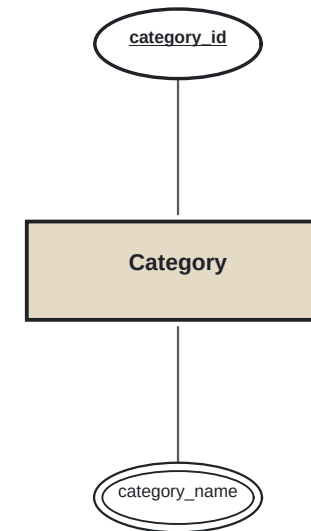
Accounts and categories

User | Category - primary key = solid underline (black), foreign key = dashed underline (grey italic), unique key = double outline



One student account. Also holds the admin.

Referenced by Product, Bid, Transaction, Wishlist, Review, Report and Notification.

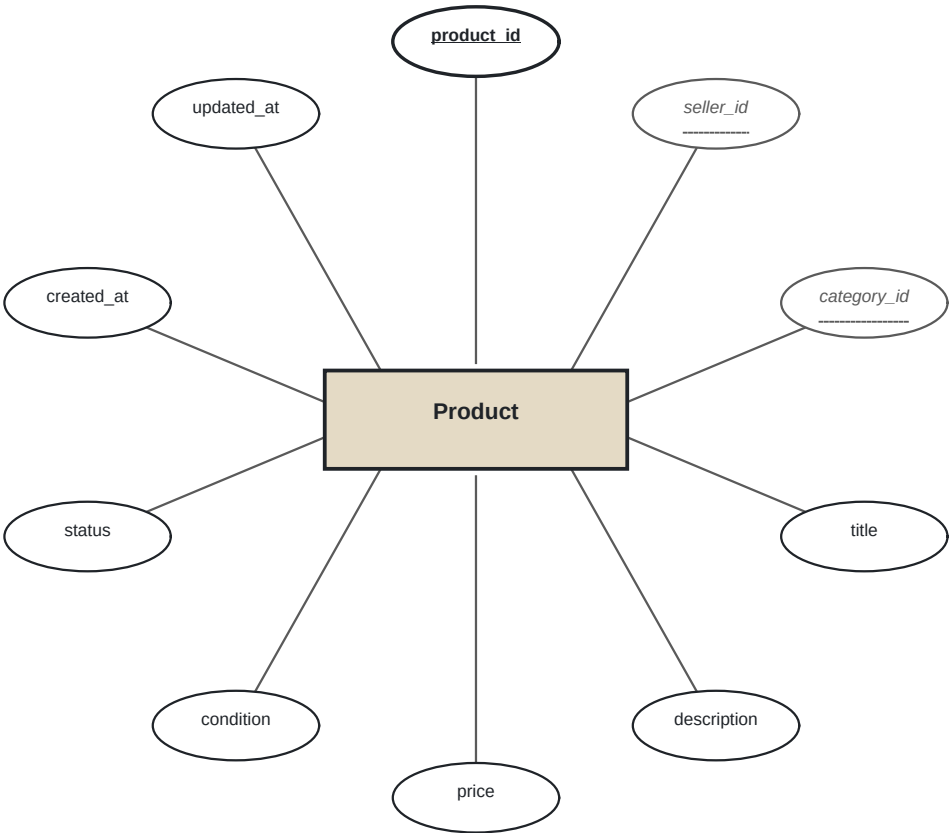


A listing category such as Books or Laptop.

Kept in its own table so a category can be renamed in one place. It has no foreign keys.

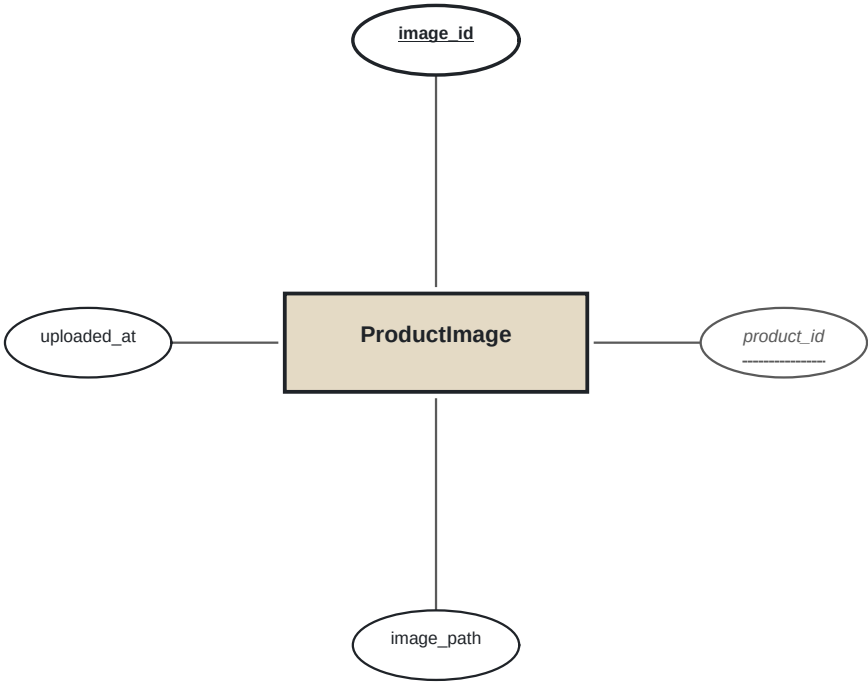
Listings

Product | ProductImage - primary key = solid underline (black), foreign key = dashed underline (grey italic), unique key = double outline



One item put up for sale.

seller_id says who is selling it, category_id which category it belongs to.

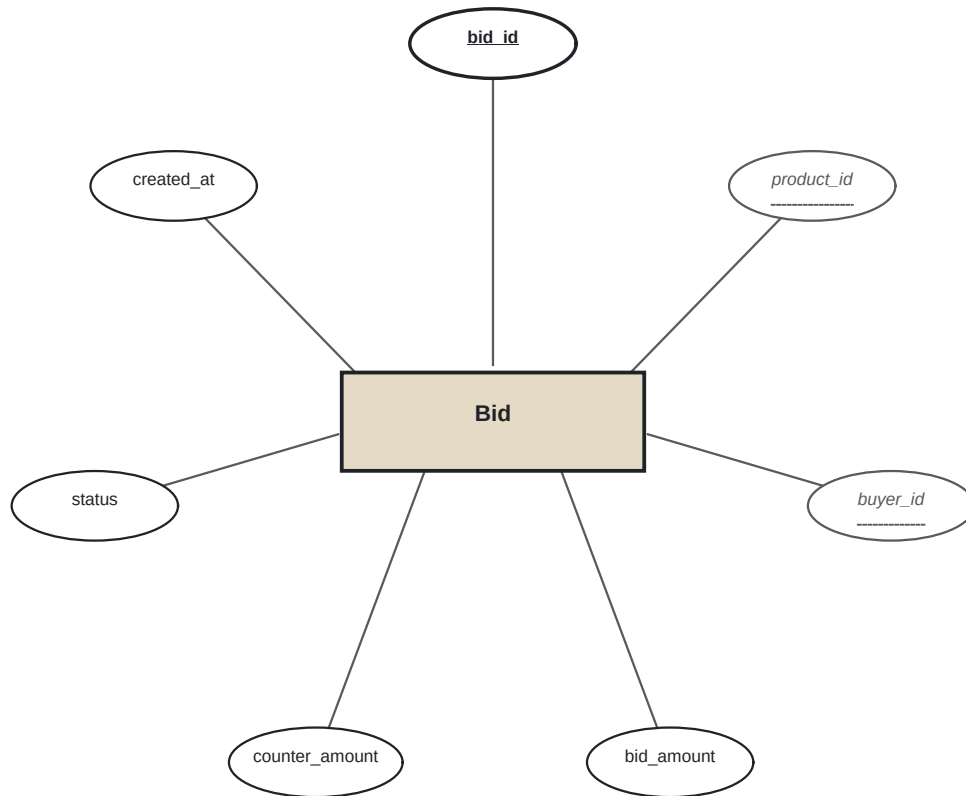


One photo of a product. A product may have several.

Kept in its own table so the number of photos is not fixed by the schema.

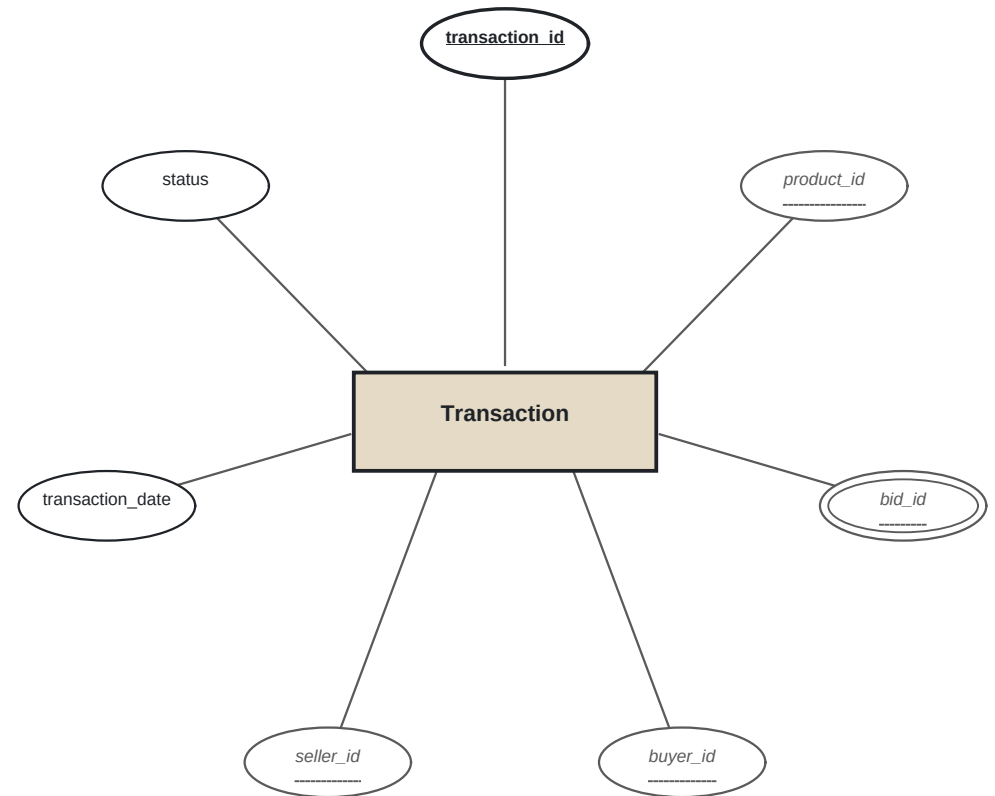
Bidding and orders

Bid | Transaction - primary key = solid underline (black), foreign key = dashed underline (grey italic), unique key = double outline



An offer a buyer makes on a product.

counter_amount holds the price the seller offers back.

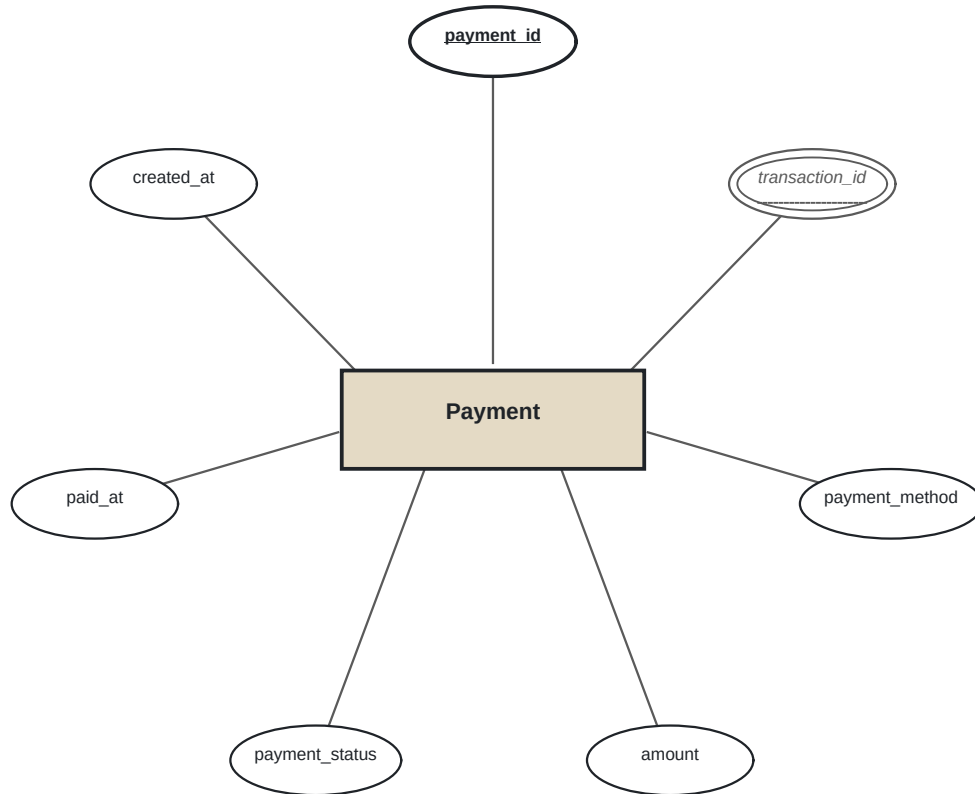


Created the moment a bid is accepted.

bid_id is unique, so one bid can produce only one order. buyer_id and seller_id are both User.

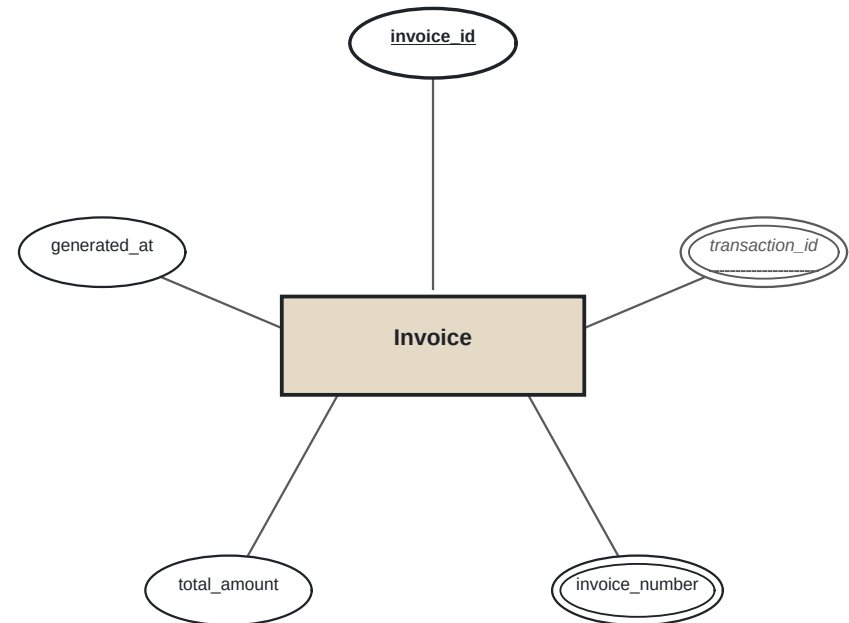
Money

Payment | Invoice - primary key = solid underline (black), foreign key = dashed underline (grey italic), unique key = double outline



How the order was paid for.

transaction_id is unique - one payment per order.

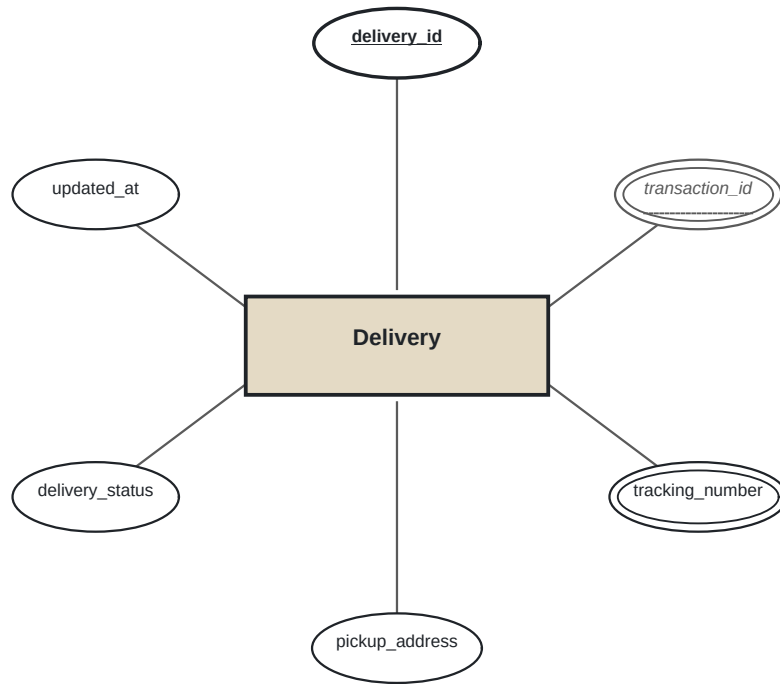


The receipt produced once an order is paid.

One invoice per order, with its own printable invoice number.

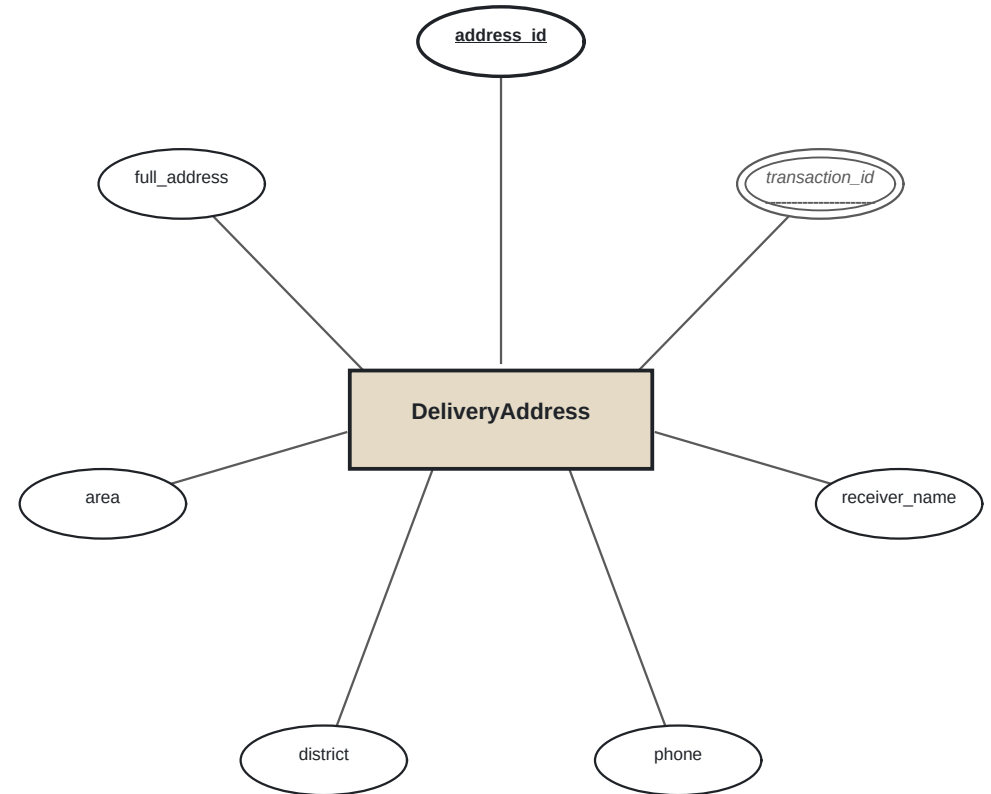
Getting the item to the buyer

Delivery | DeliveryAddress - primary key = solid underline (black), foreign key = dashed underline (grey italic), unique key = double outline



Tracks the handover of the item.

pickup_address is filled in by the seller for cash on delivery.

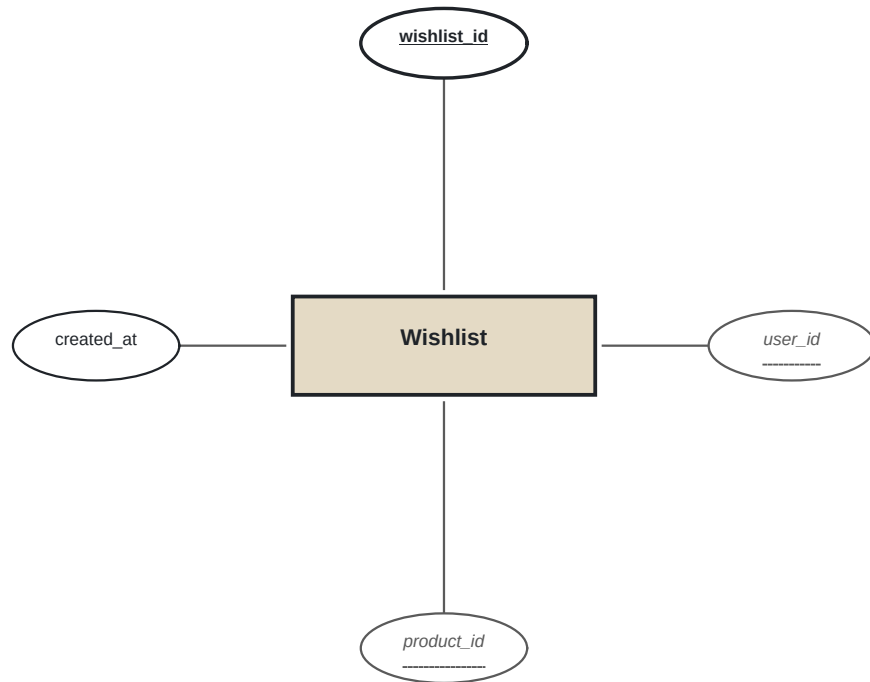


Where the buyer wants the item sent.

Split into district, area and street so addresses stay searchable.

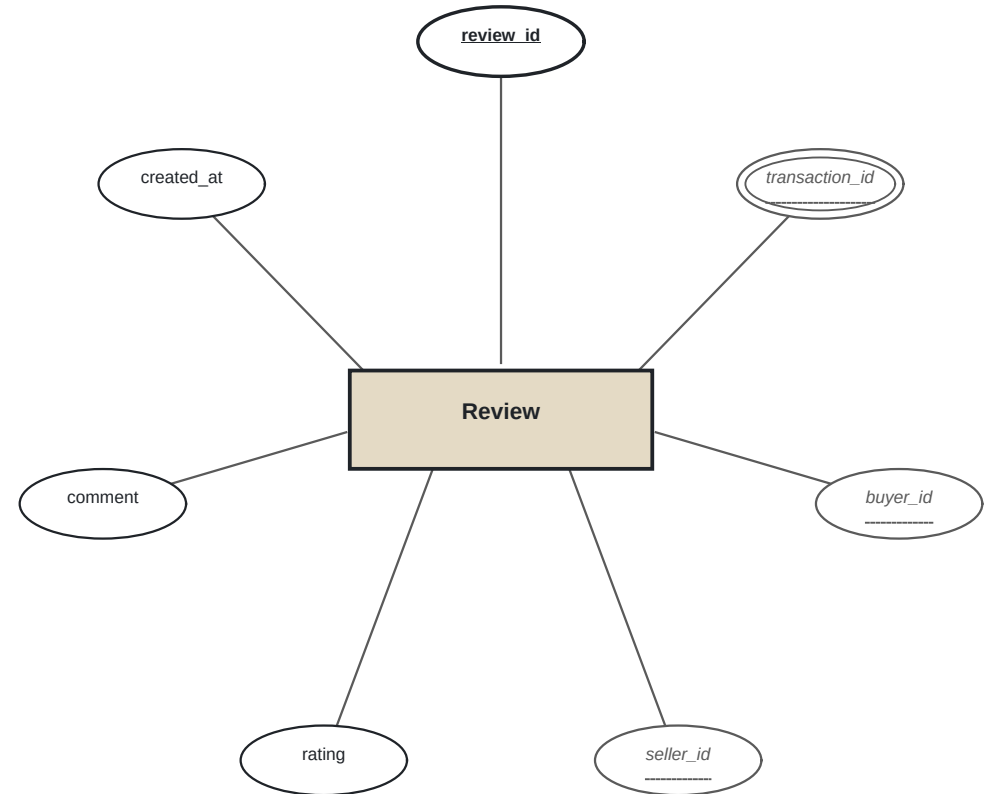
Saving and rating

Wishlist | Review - primary key = solid underline (black), foreign key = dashed underline (grey italic), unique key = double outline



A product a student saved for later.

Two foreign keys and nothing else - it exists only to join User to Product.

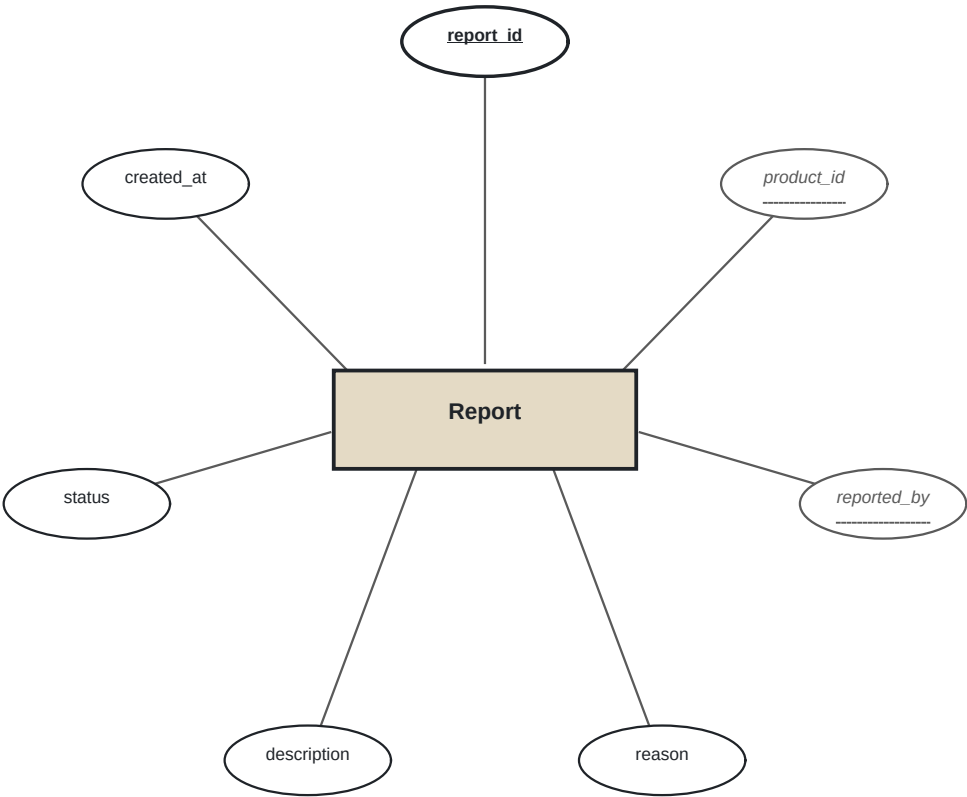


The buyer rates the seller after delivery.

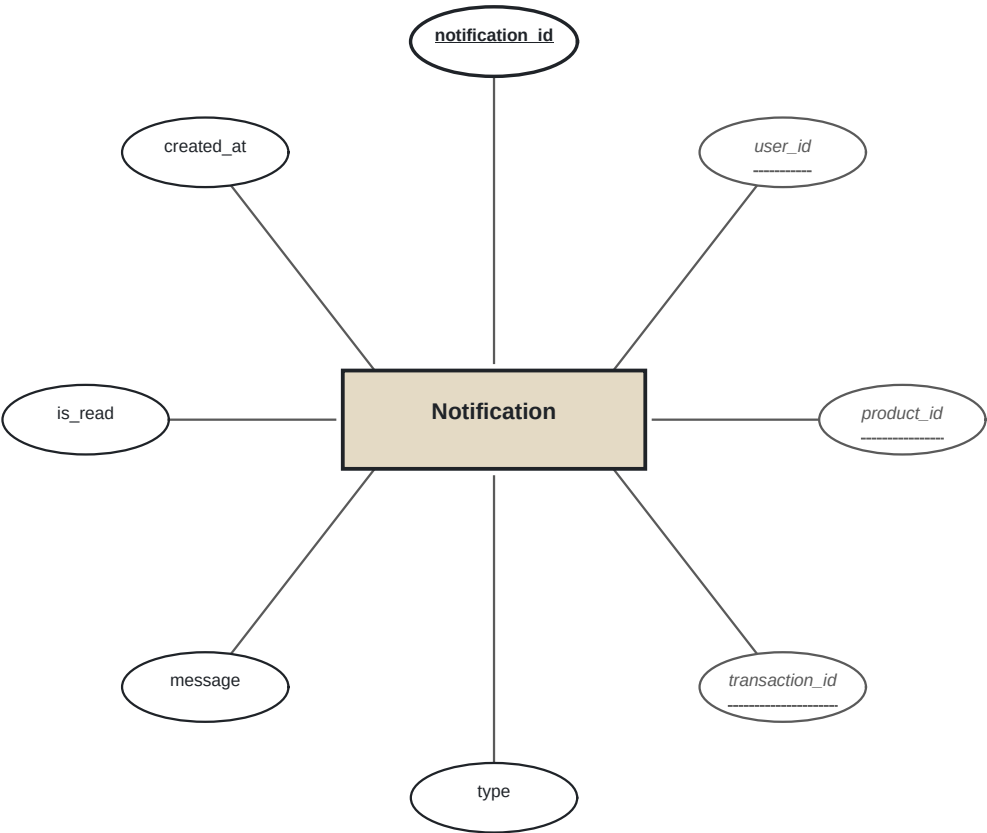
transaction_id is unique, so an order can be reviewed only once. buyer_id wrote it, seller_id is rated by it.

Moderation and alerts

Report | Notification - primary key = solid underline (black), foreign key = dashed underline (grey italic), unique key = double outline



A complaint raised against a listing.
reported_by is the student who raised it.



A message shown in the bell menu.
product_id and transaction_id may be empty; when set, clicking the notification opens that item.

All 23 foreign keys

One row per FOREIGN KEY in database.sql. Use this page to check the diagram against the schema.

Table	Column	Points at	Relationship	Type
Bid	<i>product_id</i>	Product	receives	1:N
Bid	<i>buyer_id</i>	User	places	1:N
Delivery	<i>transaction_id</i>	Transaction	handed over	1:1
DeliveryAddress	<i>transaction_id</i>	Transaction	ships to	1:1
Invoice	<i>transaction_id</i>	Transaction	billed by	1:1
Notification	<i>user_id</i>	User	notified	1:N
Notification	<i>product_id</i>	Product	about (may be empty)	1:N
Notification	<i>transaction_id</i>	Transaction	about (may be empty)	1:N
Payment	<i>transaction_id</i>	Transaction	paid by	1:1
Product	<i>seller_id</i>	User	posts	1:N
Product	<i>category_id</i>	Category	classifies	1:N
ProductImage	<i>product_id</i>	Product	has	1:N
Report	<i>product_id</i>	Product	about	1:N
Report	<i>reported_by</i>	User	files	1:N
Review	<i>transaction_id</i>	Transaction	rated by	1:1
Review	<i>buyer_id</i>	User	writes	1:N
Review	<i>seller_id</i>	User	rated	1:N
Transaction	<i>product_id</i>	Product	sold in	1:N
Transaction	<i>bid_id</i>	Bid	leads to	1:1
Transaction	<i>buyer_id</i>	User	buys	1:N
Transaction	<i>seller_id</i>	User	sells	1:N
Wishlist	<i>user_id</i>	User	saves	1:N
Wishlist	<i>product_id</i>	Product	for	1:N

Type 1:1 means the foreign key column also carries a UNIQUE index, so the parent row can have at most one child. Type 1:N means it does not, so one parent row can have many children.

User appears as the parent of nine of these, which is why it sits on the left of the overview.